Vel Tech
Rangarajan Dr. Sagunthala
R&D Institute of Science and Technology
(Deemed to be University Estd. u/s 3 of UGC Act, 1956)

Batch No: MP-AIDS-13

School of Computing

Department of Artificial Intelligence and Data Science

10214AD701- MAJOR PROJECT

Academic Year: 2025-2026

Major Project - Abstract Form

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TITLE & ABSTRACT:

Title: Hybrid Photovoltaic and Ducted Energy Integration for Dynamic EV Charging

Abstract : The growing adoption of electric vehicles (EVs) is largely driven by concerns over environmental sustainability and the rising cost of fossil fuels. However, one of the major challenges faced by EV users, especially during long-distance and intercity travel, is the limited availability of charging infrastructure. Although charging stations exist in certain locations, their sparse distribution often leads to delays and range anxiety, hindering the overall travel experience. To address this issue, this project proposes an innovative solution that integrates the Internet of Things (IoT) and Artificial Intelligence (AI) to enable automatic, on-the-go charging of EVs using renewable energy sources such as solar and wind power.

This system is designed to activate based on real-time weather conditions and battery requirements, ensuring optimal energy utilization while reducing dependency on fixed charging stations. The proposed solution not only enhances the convenience and efficiency of EV travel but also contributes significantly to reducing greenhouse gas emissions and fossil fuel consumption. By harnessing renewable energy, the system supports sustainable transportation and extends the life of EV batteries by preventing overcharging and energy waste.

Moreover, this flexible charging approach is suitable for both urban and rural environments, offering a scalable and environmentally friendly alternative for global transportation needs. Through the widespread adoption of this technology, cities can experience a notable decline in air pollution and energy inefficiency. Ultimately, this project aims to create a seamless and sustainable travel experience for EV users, aligning with global efforts to combat climate change and promote clean energy solutions.

Keywords : Electric Vehicles, Renewable Energy, IoT, AI, Smart Charging, Sustainability**.**

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